

Sample End Semester Question Paper - 2025

Subject: Database Management Systems | Code: CS302 | Semester: 3
Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define a primary key.
2. State one requirement of 2NF.
3. Name the four properties of ACID.
4. What is a deadlock?
5. What is an index?
6. Define DBMS. State two advantages over file-based systems.
7. Name the three schema levels of the ANSI-SPARC architecture.
8. What is an E-R diagram? Name its three basic symbols.
9. State one difference between DDL and DML.
10. What does the WHERE clause do in SQL?

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain database security: authentication, authorisation and integrity.
2. Explain nested queries and correlated sub-queries with examples.
3. Explain the E-R model: entities, attributes, relationships and cardinality. Convert a sample E-R diagram into tables.
4. Explain SQL with DDL, DML, DCL and TCL, giving one example statement for each.
5. Write SQL queries using WHERE, GROUP BY, HAVING and JOIN for a given sample schema.
6. Explain normalisation: 1NF, 2NF, 3NF and BCNF with a worked example.
7. Explain transaction management and ACID properties with examples.
8. Explain deadlocks: conditions, prevention, avoidance and detection. Solve a Banker's algorithm example.
9. Explain views: types, creation and applications.
10. Explain locking and concurrency control. Discuss serialisability.